

Signed order flow as a transport current: measurement, response-field theory, and memory-amplified counting noise

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We test how temporal organisation amplifies the counting variance of signed taker flow in four primary cryptocurrency markets over four years and a post-hoc four-asset extension. Raw signed volume and volume-normalised imbalance are analysed in parallel. Stratified permutation nulls separate sign persistence from magnitude clustering while preserving calendar scale changes and hour-of-week seasonality. A calibration on 14.8 million individual trades shows that the usual hourly size proxy strongly biases the absolute Fano level but leaves this ratio unchanged. Moving-block bootstrap intervals quantify uncertainty in the weekly observed/null ratio. Drift-resistant estimators give superdiffusive variance exponents 1.10–1.36, but the single-power correlation model fails in all eight series and the two-regime band is only a consistency check. We expose two closures that prevent overinterpretation: the counting variance is fixed by the correlation, and a linear fluctuation symmetry with slope $2 \langle Q_T \rangle / \text{Var}(Q_T)$ has an exact Gaussian null explanation. A response field theory and a calibrated noninteracting transport device organise these measurements without treating the device normalisation as a universal market bound. The robust departure is memory-amplified noise beyond preserved one-point statistics; the finer crossover of local exponents remains open.

I. INTRODUCTION

A market has two sides. Buyers and sellers meet through an order book, and the net signed flow of executed orders is a current in the ordinary sense: it counts something that moves from one side to the other, it fluctuates, and it responds to the conditions that drive it. This is not a metaphor we intend to defend on aesthetic grounds. It is a claim that a specific and well-developed measurement programme — the one built for transport in open, driven, mesoscopic systems — can be carried out on market data, and that carrying it out completely tells us something that measuring any one piece of it does not.

That programme has a standard shape. One measures the current, then the linear response of the system to the drive, then the noise, and then — the part that is genuinely informative and usually the hardest — the full counting statistics of the charge transferred in a finite window, including the higher cumulants and the symmetry of the distribution under reversal of the counted charge. Each layer constrains the next. Our purpose here is to do all of it on the same data set, with the same conventions, so that the layers can be compared rather than quoted in isolation.

What is already known, and what we are not claiming. We want to be precise about our position in a crowded neighbourhood, because several nearby framings are already occupied and it would be easy to appear to be claiming them.

Statistical-physics readings of markets are not new. Effective temperatures have been defined from the ratio of positive to negative returns [1], from the kinetic and potential energies of the limit order book [2], and from asymmetries in position holding times [3]; fluctuation relations have been examined for volatility cascades. Closest to our observables, Yura *et al.* [4] studied

the order book as a colloidal fluid surrounding an effective Brownian particle and reported that a fluctuation–dissipation relation *holds*, with an inner layer of short memory and an outer layer of long memory. We measure a fluctuation–dissipation ratio that is manifestly *not* constant. The two results are not in conflict — they concern different observables (the midprice particle versus the signed flow as a current) — but the difference is sharp enough that we flag it here rather than leave it for the reader to discover.

Likewise, the persistence of order flow and its coexistence with nearly martingale prices is long understood, and its microscopic origin in order splitting is by now quantitatively established [5–7]. The resolution — prices respond to the surprise in the flow rather than to the flow itself — is not ours. It is stated as a theorem in the history-dependent impact model of Taranto *et al.* [8], where $r_t = G(1)(\varepsilon_t - \hat{\varepsilon}_t) + \eta_t$ with $\hat{\varepsilon}_t$ the linear predictor of the sign, so that the price is an exact martingale by construction; the decaying propagator and the innovation representation are the same object written twice, the propagator kernel being the predictor coefficients with the sign reversed. We use this below only as a check that our conventions are right.

Two further attributions are owed, and both cut close enough that we make them before stating our own claim. First, the moments of the transferred flow are *not* unmeasured. Maitrier and Bouchaud [9] define the generalised imbalance $I_T^a = \sum_t \varepsilon_t q_t^a$ — our Q_T at $a = 1$ — and give both its variance and its higher even moments as powers of T , validated empirically to sixth order, attributing the anomalous scaling to order splitting. Our Fano growth is an algebraic rewriting of that result, and the underlying variance scaling is older still, implicit in the Hurst exponent of Lillo and Farmer [10]. Second, the log-ratio estimator of a fluctuation symmetry has al-

ready been applied to markets: Ramezani *et al.* [1] write $P(+Q, \tau)/P(-Q, \tau) = \exp(Q \Delta\beta)$ for returns, and Gao *et al.* [3] apply the same construction to position holding times. The technique is not ours; the observable is.

We should also distinguish a nearby field-theoretic treatment. Shi *et al.* [11] build a non-equilibrium field theory of the Bitcoin order book in which the net market-order discrepancy enters a continuity equation as a boundary current driving the book’s density field. That is order flow as a current, in this asset class, in non-equilibrium language. The object differs from ours: they model the density field of the book and treat flow as its boundary source, whereas we treat the signed-flow time series itself as the transport current and measure its memory kernel, response and counting statistics.

What we claim is accordingly narrower than a new observable, and we think stronger for being narrow. It is the *joint and named* characterisation. The pieces above were obtained in largely unrelated vocabularies — a propagator fitted by deconvolution, a Hurst exponent, a branching ratio, a variance-to-mean ratio, a log-ratio slope — and each was reported without reference to the others. Read as transport they are the Langreth components of one generating functional: the propagator is the retarded response, the variance-to-mean ratio is a susceptibility, the kernel-from-correlation inversion is a fluctuation–dissipation relation, and the log-ratio slope is an affinity. Measuring them on one data set with one set of conventions is what allows the three sectors of Sec. VIII — closed, explained, open — to be separated at all, and separating them is the output of the paper. The specific quantities we could not find reported for the signed flow are the joint response–counting closure, a transfer-size-preserving temporal null for counting variance, and the finite-frequency noise spectrum of the current.

The central result. Since what follows is a systematic characterisation, it is worth saying at the outset which part of it we regard as the finding and which part is scaffolding.

The finding is that temporal organisation amplifies weekly counting variance beyond what the observed one-point statistics can produce. We test this statement for both signed base volume and volume-normalised imbalance. The joint-ordering null is then decomposed: one surrogate preserves the magnitude path while permuting signs, and another preserves the sign path while permuting magnitudes. This distinguishes the dominant contribution of sign persistence from magnitude clustering without pretending that either surrogate is a structural order-book model. The counting variance is also superdiffusive under drift-resistant estimators, consistent with long memory and with the microscopic order-splitting account [5–7].

The scaffolding is everything needed to interpret that ratio. The absolute Fano level is normalisation-sensitive and can be large for an independent compound process. The informative statements are instead that the second-order sector closes, that the response is accounted for,

that the correlation carries two decay regimes, and that the observed affinity is what a Gaussian would give anyway. This is why the paper is organised along one spine: measurement, exact closure, robust temporal remainder, and calibrated model benchmark.

Why cryptocurrency, and what that costs. Three properties of these markets matter for this measurement. Public candle data carry the taker-side volume directly, so the signed flow is recoverable without proprietary feeds; trading runs continuously, so there are no overnight gaps to interrupt two-time estimates; and four years of hourly data give 3.5×10^4 observations per asset, which the counting statistics need since the higher cumulants converge slowly under heavy tails.

The cost is generality. These are venues with their own microstructure, and the order-splitting mechanism whose signature we find in the slow correlation regime was established on equity markets [5–7]. We regard the mechanism as the universal part and the venue as incidental, but that is an expectation and not something this data set can test. The statistical scaling and response statements below are about these four primary pairs; the weekly-null attribution is additionally tested on four post-hoc assets.

A dictionary between market microstructure and transport

The translation used in this paper is operational rather than metaphorical: each transport term names a quantity that can be computed from the market record. Table I also states where the analogy stops. A physicist should not infer an entropy current or a fermionic exclusion principle from the word “current”; a finance reader should not read the measured response as a causal metaorder-impact curve.

The central calculation can be read without transport vocabulary. Each hour supplies a signed number: positive when taker buyers dominate and negative when taker sellers dominate. Adding 168 consecutive hours gives one weekly net flow. Its variance across weeks is compared with synthetic histories that retain the same calendar-conditioned values but alter their temporal order. Repeating the operation after shuffling only signs or only magnitudes asks which part of that ordering carries the excess. The transport formulation becomes useful after this statistical question is clear: it supplies the counting language, exact closures, and a calibrated short-memory benchmark.

Claims are labelled by logical status throughout. A *measurement* is read from the data and carries an uncertainty or null comparison. A *closure* is an identity or model consequence and is not independent evidence. A *benchmark* shows what a specified physical model produces without asserting that the market is that model. An *open diagnostic* defines a future test but does not support a present conclusion.

TABLE I. Operational dictionary. The middle column gives the transport name; the final column states the interpretation supported by the measurement.

Market quantity	Transport language	Meaning and limit
Signed taker volume ε_t	Instantaneous current	Buyer-initiated minus seller-initiated executed volume in one candle. It is an aggregate market observable, not electric charge or entropy production.
$Q_T = \sum_{t=1}^T \varepsilon_t$	Transferred charge	Net signed volume accumulated over a declared window. It permits counting statistics, but no conservation law for a closed physical device is assumed.
$C_\varepsilon(\tau)$	Current-noise kernel	Temporal memory of signed flow. Its double sum fixes $\text{Var}(Q_T)$ exactly, so these are two views of one second-order object.
$R(\tau)$	Retarded response	Lagged covariance between later price change and present flow. It is an aggregate response diagnostic, not identified causal impact or electrical conductance.
$s(Q) = \log[P(Q)/P(-Q)]$	Fluctuation-symmetry function	A finite-window distributional asymmetry. A linear form is automatic for a Gaussian and is not by itself a fluctuation theorem.
$A_T = 2 \langle Q_T \rangle / \text{Var}(Q_T)$	Operational affinity	The Gaussian log-ratio slope. Calling it a thermodynamic force would require stationarity, microreversibility, and an appropriate large-deviation principle, none established here.
$\mathcal{F}$	Fano factor	Variance relative to a declared transfer-size quantum. Its absolute level depends on the trade-size distribution; observed/null ratios remove that arbitrary quantum.
$-C'_\varepsilon/R$	Formal FDT ratio	A derivative-based diagnostic, not a measured temperature. We derive its model form but withdraw it empirically because uncertainty is unresolved.

II. DATA AND BASE MEASUREMENT

We use hourly and four-hourly candles for four cryptocurrency pairs (BTC, ETH, BNB, SOL against USDT) over four years, 35 063 hourly and 8 766 four-hourly observations per asset. We declare two observables and carry both through the memory analysis: raw signed base volume $\varepsilon_t^{(r)} = 2V_t^{\text{taker buy}} - V_t$ and dimensionless imbalance $\varepsilon_t^{(n)} = \varepsilon_t^{(r)}/V_t$ (set to zero for a zero-volume candle). The earlier correlation and scaling figures use $\varepsilon^{(n)}$; every result below states which definition it uses. No proprietary data is required. Analysis code, numerical outputs, and data provenance are available [12].

For the central weekly-null result only, we add hourly XRP, ADA, DOGE, and AVAX as a post-hoc robustness panel selected before inspecting their decomposition outcomes. They test cross-asset generality but do not enter the original all-assets decision rule or the response and scaling fits below.

In microstructure terms, a taker removes resting liquidity: a buyer-initiated trade executes against an ask and a seller-initiated trade against a bid. The candle records their aggregate volumes, not the sign and size of every individual trade. Accordingly, $\varepsilon^{(r)}$ is a signed volume imbalance and $\varepsilon^{(n)} \in [-1, 1]$ is the fraction of that candle's volume imbalance; neither is a sequence of individually identified market orders.

We validate the candle construction against public

individual-trade archives for one complete UTC day. The sample contains 9,797,860 BTCUSDT and 4,969,357 ETHUSDT trades. Aggregation reproduces each hourly base volume, taker-buy base volume, and trade count exactly up to floating-point roundoff. This sample is used only to calibrate the transfer-size normalisation; it is not treated as an independent long-memory sample.

Three conventions matter enough to state explicitly.

Causality. Every window is trailing. This is the property on which everything else depends, and we test it directly: perturbing the future of the series must leave past features numerically unchanged.

Alignment. Flow features live on the candle index while returns are differences of log prices, so the feature of candle j corresponds to return index $j - 1$. Getting this offset wrong introduces look-ahead that no window-based test detects, because the features do not move — the labels do.

Sign of impact. The impact of ε_t is already inside the closing price of candle t , so referencing the response to $\log(\text{Close})$ measures the subsequent relaxation and returns a spurious negative impact. We reference to the opening price.

A limitation of public data, stated up front. Naviglio *et al.* [13] show that impact estimated from public market data is systematically distorted: models trained on public flow produce price paths that rise linearly rather than concavely during execution and revert too little afterwards, and they trace the problem to a misattribution

of the origin of order-flow autocorrelation. The warning applies to our data source and we do not claim to evade it. What we would argue is that it bites hardest on the quantity we are not estimating. We do not reconstruct metaorders or measure their impact; we measure the aggregate response of the price to the aggregate signed flow, together with the noise and counting statistics of that flow. Those are properties of the public series itself rather than inferences about hidden parent orders. The distinction matters for Sec. III, whose observables involve no impact estimation at all, and it should temper the reading of the response measurement in Sec. II, which is an aggregate lagged covariance and not a metaorder impact curve.

A. Memory of the flow, and why one exponent is not enough

The flow autocorrelation $C_\varepsilon(\tau)$ decays as a power law, and this is the standard point of contact with the literature: order splitting predicts long-memory signs [5], verified microscopically [6, 7]. Fitting a single exponent over 60 lags we obtain $\gamma = -0.380 \pm 0.052$ (BTC) and -0.474 ± 0.052 (ETH) at hourly resolution, broadly consistent with the reported $\gamma \approx -0.5$.

That agreement, however, needs a caveat that turns out to organise much of this paper. *The single exponent is not well defined without declaring the lag window*, because C_ε has two regimes. Fitting local slopes separately we find, for the raw hourly series,

$$a_f = 1.00 \pm 0.15, 0.87 \pm 0.08, 0.46 \pm 0.05, 1.72 \pm 0.23 \quad (1)$$

over $\tau \in [2, 10]$ for BTC, ETH, BNB and SOL respectively, against

$$a_s = 0.17 \pm 0.13, 0.22 \pm 0.09, 0.29 \pm 0.05, 0.16 \pm 0.32 \quad (2)$$

over $\tau \in [10, 50]$. The fast regime is steeper than the slow one in all four assets, though the values themselves vary considerably between assets and the slow exponents carry large errors. A single-exponent fit over a wide window returns something in between, which is why quoted values of γ are sensitive to the fitting range.

We checked whether the slow regime is an artefact of daily periodicity: a periodic mean contributes a non-decaying term to the autocorrelation and could masquerade as slow memory. Removing the seasonal component barely moves the slopes (e.g. BTC: $-1.003 \rightarrow -0.941$ fast, $-0.168 \rightarrow -0.212$ slow), so the slow regime is genuine multi-day stochastic memory rather than the drive.

B. Response and an unresolved fluctuation–dissipation diagnostic

The lagged impact $R(\tau)$ is nearly flat in τ , in sharp contrast to C_ε , which falls by roughly an order of magnitude over the same range [Fig. 1(a)]. Fitting a log-slope

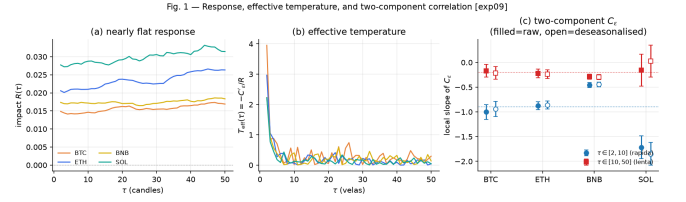


FIG. 1. Base transport measurement at hourly resolution. (a) The impact response is nearly flat in lag. (b) The raw fluctuation–dissipation diagnostic $T_{\text{eff}}(\tau)$; no inference is made from this derivative-based curve because its uncertainty is not identified and, with flat R , its shape is inherited from C'_ε . (c) Local decay exponents of C_ε in the fast and slow windows, raw (filled) and deseasonalised (open); removing the daily cycle does not move them, so the slow regime is not the drive.

to R gives values scattered around zero and small compared with γ — e.g. $b_R = 0.064 \pm 0.006$ (BTC·1h) and 0.094 ± 0.008 (ETH·1h) — though we note honestly that some series depart from flatness at resolvable significance ($b_R = -0.149 \pm 0.017$ for BNB·4h). “Flat” here means flat to within about a tenth of the decay exponent of the correlation, not flat exactly.

Flat response is what the surprise mechanism predicts [14–16]: if the price integrates the unpredictable part of the flow, the predictable part moves nothing and the lagged covariance saturates immediately. We recover it as a check on our conventions, not as a result.

One can formally construct the fluctuation–dissipation ratio

$$T_{\text{eff}}(\tau) \equiv -C'_\varepsilon(\tau)/R(\tau) \quad (3)$$

but it differentiates a noisy sample autocorrelation and divides by an estimated response. The archived point-wise curve used a finite difference and did not propagate uncertainty from either operation; its fitted slopes range from -1.02 ± 0.27 to values compatible with zero, and the proposed single-exponent closure fails in all eight series. We therefore withdraw T_{eff} as an empirical result. Figure 1(b) is retained only to show why: with a nearly flat R , the plotted shape is largely a transformed version of C'_ε , not an independently resolved thermodynamic observable.

III. COUNTING STATISTICS OF THE TRANSFERRED FLOW

We now count. Let $Q_T = \sum_{t=1}^T \varepsilon_t$ be the flow transferred in a window of T candles, and let

$$s(Q) = \ln \frac{P(Q_T = +Q)}{P(Q_T = -Q)} \quad (4)$$

be its fluctuation symmetry function. In transport, linearity of s with slope A is the signature of a driven steady state, and we follow the current fluctuation theorem of

Andrieux and Gaspard [17] in calling A the *affinity* — the conjugate thermodynamic force of the counted current, in the sense in which affinities arise from the cycle decomposition of a Schnakenberg network.

Two disclaimers on the name, both necessary. First, A is not a temperature, and we avoid that word deliberately. At least seven mutually incompatible “market temperatures” are already in print — from return log-ratios [1], holding-time asymmetries [3], order-book energies [2], Gibbs weights over strategies, volatility-cascade bath couplings, index means, and the noise parameter of minority games — and none of them is a measured fluctuation–response ratio. Adding an eighth would subtract clarity.

Second, and more seriously: Q_T is a signed particle current, not an entropy production, and the Gallavotti–Cohen symmetry is guaranteed only for the latter. Barato and Chetrite [18] give the conditions under which a *non-entropic* current inherits a large-deviation function with the Gallavotti–Cohen symmetry, and we do not establish that those conditions hold here. We therefore treat the linearity of $s(Q)$ as a phenomenological symmetry to be tested rather than a theorem to be confirmed, and the reader should read A as an operational slope with the dimensions of an affinity. Sec. III A makes the point sharper still: for a Gaussian Q_T the linearity is not even phenomenological, but algebraic.

There is also a scaling obstruction. A conventional current large-deviation principle has speed T , whereas the measured accumulated-flow variance grows as T^ν with $\nu > 1$. The natural fluctuation scale is therefore not the diffusive scale assumed by a stationary finite-correlation current. We do not establish an alternative large-deviation speed or rate function. Consequently the finite-window log ratio below is an empirical distributional statistic, not evidence that a Gallavotti–Cohen large-deviation theorem applies.

One consequence of this framing deserves flagging, because it constrains what the two halves of this paper can independently claim. Andrieux and Gaspard [19] show that the fluctuation theorem for currents fixes the response coefficients to arbitrary order from the fluctuations of the accumulated currents, reducing near equilibrium to Green–Kubo. The affinity of this section and the fluctuation–dissipation ratio of Sec. II B are therefore not logically independent measurements, and a relation between them is expected rather than surprising.

A. An identity that must not be oversold

Before reporting anything we state the caveat that governs how these results should be read. For a Gaussian Q_T with mean m_T and variance V_T ,

$$s(Q) = \frac{(Q + m_T)^2 - (Q - m_T)^2}{2V_T} = \frac{2m_T}{V_T} Q, \quad (5)$$

exactly and always. A linear fluctuation symmetry whose slope equals $2\langle Q_T \rangle / \text{Var}(Q_T)$ therefore has an exact algebraic null explanation. Observing it is not, by itself, evidence of a physical fluctuation theorem. We emphasise this because the surrounding literature makes it easy to infer more from the straight line than the measurement establishes.

Accordingly we report the closures below as consistency checks, and place the weight of the paper on the quantities that Eq. (5) cannot account for.

B. What closes

The affinity exists and is far from the sign-randomised null in all eight series. Splitting each series into four consecutive periods, the sign of A is stable across all four in BTC and ETH, and is at the noise level in BNB and SOL; the taker flow is persistently net selling in BTC, ETH and SOL over the sample. At $T = 16$ hourly candles, $A = -1.550 \pm 0.098$ (BTC) and -0.576 ± 0.074 (ETH). The measured A tracks $2\langle Q_T \rangle / \text{Var}(Q_T)$ across all windows [Fig. 2(a)] — consistent with the Gaussian identity of Sec. III A, but not proof that the distributions are Gaussian.

The second-order closure with the correlation sector is likewise clean: the counting variance reconstructed from the sample C_ε agrees with the observed one to within 4.6% in the worst case, the ratio spanning 0.954 to 1.040 over 64 window–series combinations. The counting variance adds no information beyond the autocorrelation. This is worth stating plainly because it is often assumed otherwise.

Where the data have the statistical power to resolve it, the symmetry function is close to linear. We denote its cubic coefficient by b_3 , not c_3 , to distinguish it from the third cumulant: $b_3 = 0.024 \pm 0.119$ for ETH·1h at $T = 16$, while BTC·1h gives 0.526 ± 0.240 . Standardised third and fourth cumulants do not converge uniformly to zero with T ; notably ETH becomes more skewed and heavy-tailed. We therefore define no Gaussian crossover scale and use Eq. (5) only as a null explanation for the observed linearity (Supplemental Material [20]).

C. What does not close

The robust remainder is temporal rather than a large absolute number. For reference we write the Fano normalisation as

$$\mathcal{F}(T) = \frac{\text{Var}(Q_T)}{\bar{q}^2 \langle N_T \rangle}, \quad (6)$$

where N_T is the number of trades. Candles provide only $\bar{q}_{\text{proxy}} = \text{median}_t(V_t/N_t)$, a median of hourly mean sizes, not the median of individual sizes. Under this declared proxy, $\mathcal{F}(T=1) = 905\text{--}3342$. The trade-level calibration of Sec. II gives exact median sizes smaller by factors

87.5 (BTC) and 71.6 (ETH), so the proxy is conservative for the level. It is not, however, a memory test: an independent compound-Poisson process already gives $\mathcal{F} = \langle q^2 \rangle / \bar{q}^2$, which can be large under a broad size distribution.

We isolate temporal dependence by comparing weekly windows with stratified permutation nulls. Complete Monday–Sunday UTC weeks are demeaned within each quarter-hour-of-week stratum. The primary null permutes the complete signed flow. Two attribution nulls separately permute its signs while preserving the magnitude path, or its magnitudes while preserving the sign path; all surrogates are re-centred in the same strata. Raw signed volume and normalised imbalance are analysed in parallel. The joint-null ratio is 2.2–3.4 for raw flow and 3.0–4.3 for normalised flow. Destroying sign order leaves nearly the same ratio, whereas destroying magnitude order lowers it to 1.7–2.2 and 1.5–1.9, respectively. Thus sign persistence is the dominant component, although magnitude organisation and sign–magnitude coupling remain. Moving-block bootstrap intervals over weekly sums quantify the sampling uncertainty (Supplemental Material [20]).

The predeclared all-assets screen $V_{\text{obs}} > 2Q_{0.9875}(V_{\text{null}})$ failed under its declared raw-flow quarterly joint null because BNB did not pass. We do not replace that outcome with the later normalised-flow analysis. Monthly and biweekly strata expose scale sensitivity, and cross-asset correlations preclude treating the four assets as independent replications.

The four post-hoc assets reproduce the same ordering. Their joint-null ratios span 2.08–5.67 for raw flow and 2.83–5.64 for normalised imbalance; in each asset the sign path retains more of the weekly variance than the magnitude path. This broadens the empirical panel without rewriting the failed predeclared gate.

The same remainder appears as superdiffusive growth. Fitted over the whole window grid the log-slope is 1.22 ± 0.02 to 1.37 ± 0.02 across series, against $[0.90, 1.14]$ for a simpler permutation null that preserves the marginal distribution but destroys memory.

The *local* log-slope drifts upward with window size in every series, from 1.14–1.27 at the shortest windows to 1.45–1.56 at the longest. We report the drift but do not interpret it, because it is not identified. The block variance is inflated by any drift in the per-block mean, which contributes a term growing as T^2 , and these series have such a drift: the flow carries a persistent sell-side bias and a daily seasonal component. Estimators built from vanishing-moment filters, which annihilate a constant, linear or quadratic mean while preserving the exponent, return $\nu = 1.10$ –1.36 against 1.31–1.43 for the block estimator, lower in all four assets; injecting each asset’s own measured drift onto a synthetic series of known ν inflates the block estimate by up to 0.13 and leaves the filtered ones fixed; and removing the drift by hand brings BTC from 1.42 to 1.09. What is robust is the level, $\nu > 1$ under every estimator — superdiffusive growth, and hence

memory decaying more slowly than τ^{-1} . The drift of the local exponent is treated below, where the same control is applied to a prediction that rests on it. Two further caveats attach to the long-window end. The number of non-overlapping windows falls as T grows — at $T = 256$ hourly candles only 137 remain — so those points are the noisiest. And in BNB the local slope turns over rather than continuing to rise, peaking near 1.48 at $T = 64$ and falling to 1.39 by $T = 256$ on the hourly series. We do not know whether that turnover is a third regime or the small-sample noise of the last few points, and we do not have the statistics to decide. The conclusion does not rely on comparing unlike absolute Fano normalisations: the observed chronology contributes a factor beyond the size-preserving null, while the correlation and accumulated variance are two exactly linked views of the same second-order memory.

IV. CLOCKS: WHAT AGES AFTER A SHOCK, WHAT DOES NOT, AND WHAT ONLY APPEARS TO

Systems with slow dynamics often age: correlation functions after a quench depend on the time elapsed since the quench, not only on the lag. We tested this directly, defining shocks as returns exceeding the 99.5th percentile in absolute value and building an ensemble of 240 hourly post-shock trajectories.

The outcome of this section is easier to follow if we state it first. Evidence for an Omori-like relaxation of the mean is weak, and the apparent ageing of the correlation is reproduced by a placebo. The latter is a property of the estimator rather than of the market: dividing a post-event signal by a quantity measured before the event manufactures ageing by construction. Identifying that mechanism and the null that exposes it is the reliable result of this section.

The *mean* normalised absolute return is compatible with an age-dependent profile of Omori form. We fit the *amplitude* exponent β in $\sigma(t) \sim t^{-\beta}$ — not the rate exponent p in $n(t) = K(t + \tau)^{-p}$, and the distinction matters because the two are routinely conflated in this literature while being related by $p = \alpha\beta$ with α the tail index [21]. Our conservative fit, excluding the first few candles where the fast relaxation dominates, gives $\beta = 0.61 \pm 1.01$; the uncertainty spans zero. Moreover, six of the ten asset-resolution fits terminate at an optimiser bound, including several with orders-of-magnitude uncertainties. We therefore report these fits as descriptive diagnostics and make no resolved claim for a universal Omori exponent.

It is worth being explicit about where that sits. Published amplitude exponents for equities cluster at $\beta \approx 0.22$ –0.32 [21, 22], so our point estimate is high by a factor of two to three, although not resolved from them at this uncertainty. Read instead as a rate exponent it would be unremarkable, falling in the mode of the distribution over crashes reported by Selçuk [23]. The consistency check

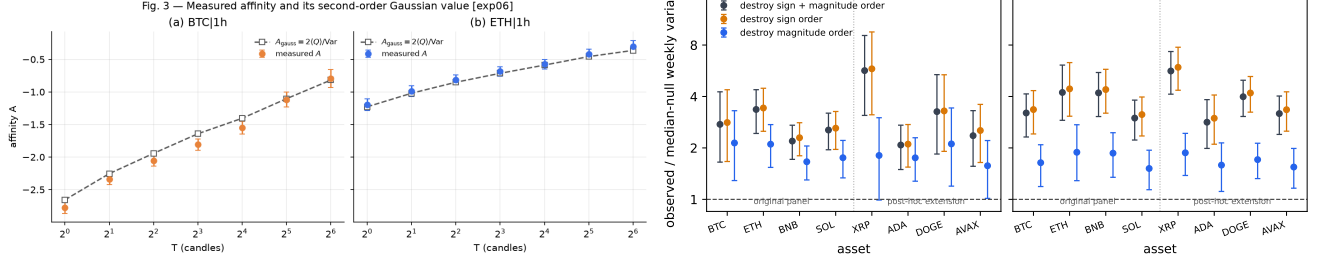


FIG. 2. Counting statistics. Left: measured affinity against the Gaussian value $2\langle Q_T \rangle / \text{Var}(Q_T)$; agreement is consistent with Eq. (5) and does not establish a fluctuation theorem. Right: weekly observed/null variance ratios for the joint, sign-order, and magnitude-order nulls under both flow definitions. Bars are eight-week moving-block bootstrap intervals; the transfer quantum cancels.

available here is $p = \alpha\beta$: with a tail index of order 3, an amplitude exponent near 0.2 would follow from a rate exponent near 0.6. We report the amplitude fit because that is the quantity we measured, and we do not convert.

The correlation time $\tau_c(t_w)$ also appears to grow with age, monotonically, with a Spearman coefficient of +1.00 across age bins.

That second result is an artefact, and the way we caught it generalises. Placing shocks at *random* times and repeating the whole analysis, the null ensemble returns a median Spearman coefficient of +1.00 as well [Fig. 3]. The cause is the pre-event normalisation: dividing by the volatility measured just before the event injects a slow common mode, because the volatility drifts away from its pre-event value as age increases, and this manufactures apparent ageing for *any* clock, including a meaningless one. The observed growth is statistically indistinguishable from that artefact, so we report no ageing in the correlation.

We state the resulting rule in general form, because the construction that produces it is not unusual: *in any event-conditioned two-time analysis, normalising by a pre-event quantity fabricates ageing, and a null built from randomly placed events is mandatory rather than optional*. The null costs one re-run of the same pipeline, which is a small price for a result that would otherwise be a discovery.

Why we think this generalises. The mechanism is not specific to markets, and versions of each half of it are established elsewhere. That normalising by a noisy denominator biases an estimator is old [24, 25]; the sharpest modern statement is that the bias is controlled by the correlation between numerator and denominator [26], which in our geometry decays with t_w and so generates a profile in age rather than a constant offset. That conditioning on an extreme value inflates a subsequently measured correlation is the Forbes–Rigobon effect [27, 28], where an event-selected variance in the denominator changes a measured cross-market correlation from 0.45 to 0.81 with no change in the data-generating process; what we describe is its two-time analogue. And the scaling argument itself has been made exactly, in another system: for

a granular fluid in the homogeneous cooling state, Brey *et al.* [29] show that normalised two-time correlation functions decay more slowly the later the measurement begins, purely because the overall scale of the signal decays self-similarly, with no memory whatever — and that the effect disappears in properly rescaled variables.

Two alternative explanations must be excluded rather than assumed away. The sample autocorrelation is biased in finite samples [30], and the resulting bias in a fitted decay time grows with record length — but that bias is negative, depends on the number of points rather than on the age, and is present in our randomly placed null as well, so it cannot produce a difference between the two [31]. Separately, a time-averaged correlation function computed from a trajectory shorter than the longest relaxation time shows apparent ageing on its own [32]; our placebo controls for this for the same reason.

Scope of the claim. We want to be careful about how far this reaches. What we have shown is that *in our data*, with our normalisation, the measured growth of τ_c with age is indistinguishable from what randomly placed events produce. That is a statement about this measurement, and it does not by itself invalidate any other.

It does, however, apply to a construction used elsewhere. The same object — a σ -normalised two-time correlator built on event-conditioned windows, with the waiting time counted from the conditioning event — underlies reports of ageing in aftershock sequences and, subsequently, in a currency crash [33, 34]. We are not aware of a published null of this kind for those analyses. Whether the effect reported there survives such a null is a question we cannot settle with market data and do not attempt to settle here; we raise it because the check is cheap and, on our data, decisive. The normalisation convention $C(t, t_w)/C(t_w, t_w)$ is standard in the ageing literature, and Brey *et al.* [29] show analytically that this convention alone produces full ageing in a system with no memory, which suggests the question is worth asking generally rather than only here.

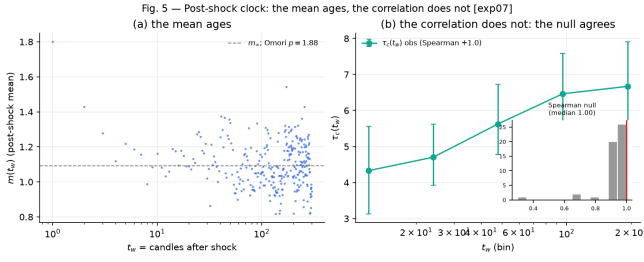


FIG. 3. Post-shock behaviour. (a) The mean flow relaxes with age. (b) The correlation time appears to grow with age, but randomly placed shocks reproduce the same monotone growth (inset), identifying it as an artefact of pre-event normalisation.

V. METHODS AND STATISTICAL CAVEATS

Three inference problems arose that we think are worth reporting on their own, because each of them produced, at some stage, a result we would otherwise have believed.

Predictive comparison is miscalibrated under heavy tails, unevenly. Comparing forecast accuracy with the standard Diebold–Mariano test [35] using Newey–West standard errors [36], we generated the null distribution by permutation through the identical pipeline. The empirical rejection rate at nominal 5% is 4–6% for BTC but 16–18% for ETH, with a null standard deviation of about 2.1 instead of 1.0. A nominal $p < 0.05$ on the heavier-tailed asset is worth roughly 0.17. The practical consequence is that nominal p values must not be compared across assets as though they measured the same thing.

Scanning scales without maxT manufactures findings. Sweeping 144 configurations (assets $\times$ intervals $\times$ windows $\times$ horizons $\times$ baselines) and applying the Westfall–Young maxT correction [37], the best cell had a nominal p of 2×10^{-4} — and yet the maximum statistic generated by chance over those same 144 cells had a mean of 3.775, above the observed maximum of 3.667. The global p was 0.43. Only five cells passed $p < 0.05$ where 7.2 would be expected by chance. Without the correction this would have been published as a discovery. Bonferroni is not an adequate substitute, since the cells share data and are far from independent.

Recalibration is not information, and correlation is not evidence. A model $\log(\text{rv}) \sim a + b \log \sigma + cX$ beats raw σ even when X is pure noise, because it re-fits level and slope; the contrast that decides must always be recalibrated-without- X against recalibrated-with- X . Relatedly, a pure AR(1) noise feature with no relation to the data correlated with out-of-sample $\log(\text{rv})$ at +0.21 (BTC) and +0.32 (ETH) — higher than the real feature. Spurious correlation between persistent series is the rule, not the exception.

Negative results need positive controls. Surrogate and permutation nulls demonstrate that a pipeline does not manufacture signal. None of them demonstrates that it

can see a signal. Every negative result reported here is accompanied by a positive control on synthetic data carrying the effect by construction, without which a null is ambiguous between “no effect” and “broken code”. This is the standard requirement of assay sensitivity [38], transposed to a computational pipeline as an injection–recovery test; we note it because retrospective power computed from an observed effect size is not a valid substitute [39].

The memory exponent is estimator-dependent, and the disagreement is itself the measurement. Lillo and Farmer [10] obtain $\gamma = 0.39$ by log–log regression on the sample autocorrelation and 0.61 by periodogram on the same data, and state that the sample autocorrelation is a poor estimator of γ . We therefore computed γ four ways — log–log regression on C_ε , a log-periodogram (GPH) regression, detrended fluctuation analysis, and the wavelet log-scale diagram of Abry and Veitch [40, 41]. The last plots $\log_2 \langle d_j^2 \rangle$ against octave j , where d_j are Daubechies detail coefficients; for a process with $C_\varepsilon(\tau) \sim \tau^{-\gamma}$ the diagram is a straight line of slope $\alpha = 1 - \gamma$. Two properties make it the natural arbiter here. A Daubechies filter with $L/2$ vanishing moments annihilates polynomial trends of degree below $L/2$ [42], so a drifting mean cannot masquerade as memory; and the details are close to decorrelated across octaves for self-similar input [43], so the fit returns an error bar and a goodness-of-fit for the power law it assumes — which the other three do not.

All four were calibrated against series with γ known by construction. The choice of calibration series is not a detail: an estimator that reads the low-frequency end of the spectrum cannot be validated against a generator whose autocovariance is matched only at short lags, since the approximation error then sits precisely in the band being measured. We therefore synthesise exact fractional Gaussian noise by the circulant method of Davies and Harte [44], whose spectral factorisation is exact for the fGn autocovariance and needs no eigenvalue truncation. Over $\gamma \in [0.4, 0.8]$ with 4×10^5 samples, the mean absolute errors are 0.011 (wavelet), 0.023 (DFA), 0.026 (regression) and 0.068 (GPH). No estimator is grossly biased and the wavelet is the most accurate, so the disagreement seen on real data cannot be attributed to any one of them being broken.

On our data the four spread by 0.23 (BNB) to 0.53 (SOL), an order of magnitude beyond what the calibration allows for estimator noise. The wavelet diagnostic supplies the reason: the log-scale diagram is not straight. A single-slope weighted fit over $j \geq 3$ returns χ^2 per degree of freedom of 3.7, 1.9, 3.1 and 6.1 for BTC, ETH, BNB and SOL, rejecting the single-power-law model outright. There is therefore no single γ for the four estimators to agree on: each returns a different weighted average over scales, according to where in the spectrum it places its weight. This is the same conclusion the two-window fit of Sec. II A reaches from the autocorrelation, obtained through a different estimator and with an ex-

explicit goodness-of-fit rather than by comparing two fitted ranges. The values in Sec. II A should accordingly be read as a band of roughly ± 0.1 rather than as three-decimal measurements, and the same caution applies to any published exponent quoted from a single estimator without the goodness-of-fit of the power law it presumes. The resolvable range is set by the record: 3.5×10^4 hourly observations give about nine usable octaves, and beyond $j = 8$ fewer than 130 coefficients survive, so the slow component sits at the edge of what these series can resolve at all.

The exponent is also not stable across data vintages: restricting the same series to its most recent two years gives $\gamma = 0.610 \pm 0.097$ for BTC against 0.380 ± 0.052 over four years. We report the four-year values throughout and flag the instability rather than selecting the vintage that agrees best with the literature.

Finite-sample autocorrelation bias cannot explain the apparent ageing. A second mechanism could in principle produce the artefact of Sec. IV: the sample autocorrelation is biased in finite samples [30], and a fitted correlation time therefore grows with the number of points available. We quantified it on AR(1) processes with known ρ : at $\rho = 0.95$ the fitted τ recovers 32% of its true value at $n = 50$ and 97% at $n = 1600$, so the effect is large enough to matter. It cannot act here, however, because the ensemble size entering each correlation estimate is the number of shocks and is therefore constant in the age t_w — we verified this directly, with a relative range across age bins of exactly zero for all four assets. A bias that does not vary with t_w cannot generate a profile in t_w . It is also present in the randomly placed null, which controls for it in any case.

VI. RESPONSE-FIELD THEORY

We now encode the measured second-order structure in a minimal stochastic field theory. The purpose is organisational, not microscopic: the action states which relations follow once the correlation kernel and innovation coupling are fixed, and therefore which observations cannot count as new evidence. Readers from finance may view the construction as a compact generator of covariances and cumulants; readers from physics should view it as a classical MSRJD theory, not as a quantum Keldysh contour for noncommuting market operators.

A. The action

We use two fields, the signed flow $\varepsilon(t)$ and the log-price $p(t)$, with conjugate response fields $\tilde{\varepsilon}$ and $\tilde{p}$, and statistical weight $\propto e^{-S}$ with $S = S_\varepsilon + S_p$.

B. Flow sector

The flow is a Gaussian field with a periodically modulated mean and a stationary memory kernel,

$$S_\varepsilon = \int dt \tilde{\varepsilon} [\varepsilon - \mu(\phi(t))] - \frac{1}{2} \iint dt dt' \tilde{\varepsilon}(t) K(t-t') \tilde{\varepsilon}(t'), \quad (7)$$

with $\phi(t)$ the daily phase. Integrating out $\tilde{\varepsilon}$ along the imaginary direction returns the Gaussian weight $\exp\{-\frac{1}{2} \iint (\varepsilon - \mu) K^{-1} (\varepsilon - \mu)\}$, so that

$$G_{\varepsilon\varepsilon}^K(t, t') = \langle \varepsilon(t) \varepsilon(t') \rangle_c = K(t-t') = C_\varepsilon(t-t'), \quad \langle \varepsilon \tilde{\varepsilon} \rangle = \delta. \quad (8)$$

We deliberately do *not* postulate a dynamics for ε . The kernel is measured, not derived, and the microscopic origin of its long-memory component is independently established as order splitting [5–7]. Treating the flow as a prescribed non-Markovian bath keeps the model honest about what it explains: all predictive content is pushed into the price sector and the counting sector, where it can be tested against observables the kernel was not fitted to.

C. Price sector

The price integrates the *innovation* of the flow rather than the flow itself,

$$S_p = \int dt \tilde{p} [\partial_t p - \lambda w] - \frac{D_p}{2} \int dt \tilde{p}^2, \quad (9)$$

where D_p is the variance of price noise unrelated to flow and w is defined as follows. With $S_\varepsilon(\omega) = \sum_k C_\varepsilon(k) e^{-i\omega k}$ the flow spectral density, positivity and $\int \ln S_\varepsilon < \infty$ give the canonical Wiener–Hopf factorisation $S_\varepsilon(\omega) = \sigma_w^2 |h(e^{-i\omega})|^2$ with $h(z) = \sum_{k \geq 0} h_k z^k$ causal, minimum-phase and $h_0 = 1$. Equivalently one has the Wold representation [45, 46]

$$\varepsilon_t - \mu_t = \sum_{k \geq 0} h_k w_{t-k}, \quad \langle w_t w_s \rangle = \sigma_w^2 \delta_{ts}, \quad (10)$$

which defines the innovation w_t as the part of ε_t not linearly predictable from its own past. The whitening filter $g = h^{-1}$ is causal, so w_t is measurable with respect to the past and the coupling in Eq. (9) respects causality, as it must.

Equations (7)–(9) are the entire model. Everything below is a consequence.

D. Why the response is flat

Proposition 1 (Flat impact). *Under Eqs. (9)–(10), the lagged impact*

$$R(\tau) \equiv \frac{\text{cov}(p_{t+\tau} - p_t, \varepsilon_t)}{\text{Var}(\varepsilon)} \quad (11)$$

is independent of τ for all $\tau \geq 1$, with $R(\tau) = R_0 = \lambda \sigma_w^2 / \sigma_\varepsilon^2$.

Proof. From Eq. (9), $p_{t+\tau} - p_t = \lambda \sum_{s=t}^{t+\tau-1} w_s$ plus noise uncorrelated with ε . Equation (10) gives $\text{cov}(w_s, \varepsilon_t) = \sigma_w^2 h_{t-s} \mathbf{1}\{s \leq t\}$. In the sum s runs over $\{t, \dots, t+\tau-1\}$, so only $s = t$ survives, contributing $\sigma_w^2 h_0 = \sigma_w^2$. Hence $\text{cov} = \lambda \sigma_w^2$ for every $\tau \geq 1$, independent of τ . $\square$

The mechanism — the price responds to the surprise in the flow, so the predictable part of a persistent flow moves no price and prices stay nearly martingale — is not ours. It is the standard resolution of the persistence-versus-martingale puzzle [14–16], and it also removes the need for a decaying propagator to reconcile correlated flow with uncorrelated returns [47, 48]. We recover it here as a check that the bookkeeping in Eqs. (7)–(9) is right. Its value is what it enables next: with R pinned flat, the fluctuation–dissipation ratio has no shape freedom left.

Remark 1. *The measured response is flat to within about a tenth of the decay exponent of the correlation, not flat exactly; some series depart from flatness at resolvable significance [12]. Proposition 1 is the idealisation, and the residual slope is a measure of how far real flow departs from the pure-innovation coupling.*

E. A formal fluctuation–dissipation ratio

Define formally $T_{\text{eff}}(\tau) \equiv -C'_\varepsilon(\tau)/R(\tau)$. By Proposition 1 the denominator is constant, so

$$T_{\text{eff}}(\tau) = -C'_\varepsilon(\tau)/R_0, \quad (12)$$

and T_{eff} inherits its entire shape from the flow autocorrelation. A single scalar “violation of the fluctuation–dissipation theorem” is thus one point of a curve that Eq. (12) determines completely. In systems with slow dynamics such ratios define effective temperatures with genuine thermodynamic content [49]; here the ratio is fixed by the correlation function and carries no independent information.

With the measured two-component kernel,

$$C_\varepsilon(\tau) \simeq A_f \tau^{-a_f} + A_s \tau^{-a_s}, \quad (13)$$

Eq. (12) becomes

$$T_{\text{eff}}(\tau) R_0 = a_f A_f \tau^{-(1+a_f)} + a_s A_s \tau^{-(1+a_s)}. \quad (14)$$

The exponents in this formal expression are not free parameters: they are $1 + a_f$ and $1 + a_s$. Empirically testing the expression is harder than writing it, because differentiating a noisy sample correlation and dividing by an estimated response requires joint uncertainty propagation. The archived finite-difference diagnostic did not provide that propagation and failed its proposed single-exponent form in all eight series. We therefore retain Eqs. (12)–(14) as model identities and make no empirical T_{eff} claim [12].

F. The counting sector

Let $Q_T = \sum_{t=1}^T \varepsilon_t$ be the flow transferred in a window of T candles. Introducing the counting field χ conjugate to Q_T — the object carrying opposite signs on the two contour branches, as in the counting statistics of mesoscopic transport [50–53] — the cumulant generating function is $F_T(\chi) = \ln \langle e^{\chi Q_T} \rangle$. For the Gaussian action (7) it truncates exactly at second order,

$$F_T(\chi) = \chi m_T + \frac{1}{2} \chi^2 V_T, \quad V_T = \text{Var}(Q_T) = \sum_{t,t'=1}^T C_\varepsilon(t-t'), \quad (15)$$

so every cumulant κ_n with $n \geq 3$ vanishes identically. The relation $V_T = \sum_{t,t'} C_\varepsilon(t-t')$ is an exact sample-covariance identity. Its 4% numerical reconstruction error over 64 window–series combinations is a pipeline check, not independent empirical confirmation [12].

1. An identity, not a fluctuation theorem

Proposition 2 (Gaussian affinity). *For Gaussian Q_T with mean m_T and variance V_T , the fluctuation symmetry function is exactly linear,*

$$s(Q) \equiv \ln \frac{P(Q)}{P(-Q)} = A_T Q, \quad A_T = \frac{2m_T}{V_T} = \frac{2\langle Q_T \rangle}{\text{Var}(Q_T)}, \quad (16)$$

and the Gallavotti–Cohen symmetry $F_T(\chi) = F_T(-A_T - \chi)$ holds identically.

Proof. From $P(Q) \propto e^{-(Q-m_T)^2/2V_T}$, $s(Q) = [(Q+m_T)^2 - (Q-m_T)^2]/2V_T = 2m_T Q/V_T$. For the symmetry, substitute $\chi \rightarrow -A_T - \chi$ in Eq. (15) with $A_T = 2m_T/V_T$, and use $V_T A_T^2/2 = 2m_T^2/V_T$ and $V_T A_T \chi = 2m_T \chi$:

$$F_T(-A_T - \chi) = -A_T m_T - \chi m_T + \frac{V_T}{2} (A_T^2 + 2A_T \chi + \chi^2) = F_T(\chi). \quad \square$$

Remark 2 (The point of the paper’s negative half). *Proposition 2 holds for any Gaussian variable. A measured linear $s(Q)$ whose slope equals $2\langle Q \rangle / \text{Var}(Q)$ therefore does not, by itself, establish more than consistency with this Gaussian null, and in particular it is **not sufficient evidence** of a nontrivial fluctuation theorem in the sense of Gallavotti–Cohen or Lebowitz–Spohn [54, 55]. Given how naturally such a measurement invites the opposite reading, we regard stating this clearly as part of the result. The informative content must be sought in resolved departures, including the cubic coefficient b_3 of $s(Q)$, the higher cumulants, and normalisation-free null comparisons.*

Remark 3 (A second reason not to call this a fluctuation theorem). *There is a structural obstruction independent of Gaussianity. The Gallavotti–Cohen symmetry is guaranteed for the entropy production; Q_T here*

is a signed particle current, which is not that. Barato and Chetrite [18] establish the conditions under which a non-entropic current inherits a large-deviation function with the Gallavotti–Cohen symmetry, and those conditions are restrictive and not established for the process studied here. Proposition 2 shows the symmetry holds trivially in the Gaussian sector; outside it, nothing guarantees the symmetry at all, which is one more reason to read the linearity of $s(Q)$ as an empirical regularity to be tested rather than a law to be confirmed.

The superdiffusive variance adds a separate caution: a conventional current large-deviation principle has speed T , while $V_T \propto T^\nu$ with $\nu > 1$. No alternative speed or rate function is established here.

Remark 4 (The affinity and the response are not independent). Andrieux and Gaspard [19] show that a fluctuation theorem for currents determines the response coefficients to arbitrary order from the fluctuations of the accumulated currents, reducing to Green–Kubo near equilibrium. The affinity of this section and the effective temperature of Sec. VIE are therefore expected to be related rather than independently informative, and we treat their consistency as a check on the framework rather than as two separate results.

2. Affinity dilution and a crossover prediction

For a single power-law kernel $C_\varepsilon(\tau) = A\tau^{-a}$ with $0 < a < 1$, the double sum in Eq. (15) gives for $T \gg 1$

$$V_T \simeq 2A \int_0^T (T-k)k^{-a} dk = \frac{2A}{(1-a)(2-a)} T^{2-a}, \quad (17)$$

so the counting variance is superdiffusive with exponent $\nu = 2 - a$. At the marginal point $a = 1$ the leading form is $T \log T$; for $a > 1$ the correlation is summable and the asymptotic variance is diffusive, $V_T \propto T$. This domain distinction matters for BTC and SOL, whose fast-regime point estimates are at or above one. Since $m_T = \bar{\mu}T$ is strictly linear, Proposition 2 forces the affinity to dilute with observation scale,

$$A_T = \frac{2\bar{\mu}T}{V_T} \propto T^{a-1}, \quad (18)$$

decreasing for every $a < 1$: memory dilutes the bias. The scale-free version of the statement requires no power-law assumption at all and is exact,

$$\frac{A_{T_2}}{A_{T_1}} = \frac{T_2}{T_1} \cdot \frac{V_{T_1}}{V_{T_2}}. \quad (19)$$

Tested against the measured variances, Eq. (19) predicts A_{16}/A_1 to within 0.5–13% for the two assets whose affinity is stable across the sample, and fails by 43% and 132% for the two whose affinity is at the noise level [Fig. 4(b)] — failing, that is, exactly where the quantity being predicted is not well determined.

A finite-scale consistency check. With the two-component kernel (13), Eq. (17) cannot hold with a single ν . Short windows sample the fast regime and long windows the slow one, so the local logarithmic slope of V_T must drift,

$$\nu_{\text{loc}}(T) : 2 - a_f \longrightarrow 2 - a_s \quad \text{as } T \text{ increases.} \quad (20)$$

For a strict sum of two power laws the drift is monotone, since ν_{loc} is a weight-average of the two exponents with the slow weight increasing in T . We state the prediction in the weaker form — direction and band — for two reasons: the empirical kernel is only approximately a two-component power law, and the strict form is the one the data will be seen below to strain. This is not an independent test: C_ε and V_T are linked by the exact finite-sample second-order identity. Nor is the interval a quantitative two-component prediction, because the fitted regime amplitudes and crossover scale were not propagated. It is only a consistency band. A single-component bath predicts constant ν_{loc} , and would be excluded by a systematic drift *provided the drift is a property of the correlations and not of the estimator*. That proviso turns out to carry the whole test.

Applying the correct domain asset by asset gives the point-estimate bands [1.00, 1.83] (BTC), [1.13, 1.78] (ETH), [1.54, 1.71] (BNB), and [1.00, 1.84] (SOL). The observed block exponents for BNB, 1.37 at both hourly and four-hourly resolution, lie outside its band. Thus even this weak consistency check fails in one of four assets and cannot be advertised as a universal closure. The observed local slope drifts upward over much of $T = 1 \dots 256$, from 1.14–1.27 at the short end to 1.45–1.56 at the long end [Fig. 4(a)]. The drift is not monotone everywhere: in BNB the local slope peaks near 1.48 at $T = 64$ and falls back to 1.39 by $T = 256$ on the hourly series, with the same turnover on the four-hourly one.

A drifting mean forges the same signature. V_T is estimated as the sample variance of Q_T over non-overlapping blocks, so any drift in the per-block mean $\langle Q_T \rangle$ enters it. A deterministic mean that varies across blocks contributes a term growing as T^2 — steeper than anything the kernel predicts, and biased in the same direction as the effect sought. This is not hypothetical for these series: the taker flow carries a persistent sell-side bias over the four-year record and modulates with the daily cycle [12]. The prediction and its confounder therefore have the same observable signature, and separating them requires an estimator of ν that is blind to the mean.

The Haar difference between adjacent blocks supplies one. With $d_T = Q_T^{(2)} - Q_T^{(1)}$ one has identically $\text{Var}(d_T) = 4V_T - V_{2T}$, so for a pure power law $V_T = AT^\nu$,

$$\text{Var}(d_T) = A(4 - 2^\nu) T^\nu, \quad (21)$$

the *same* exponent with a constant mean cancelled exactly. Applied directly to the flow, raising the number of vanishing moments raises the degree of the annihilated polynomial: db1 removes a constant mean, db2 a linear

TABLE II. Global scaling exponent ν of the counting variance over $T = 8 \dots 256$, by four estimators. V_T is the block variance; m is the number of vanishing moments, so $m = 1$ (Haar) is blind to a constant mean, $m = 2$ to a linear drift and $m = 3$ to a quadratic one. The final column is the inflation of the block estimate produced by injecting that asset's own measured mean drift onto a synthetic series of known ν ; the Haar estimate moves by less than 0.01 in every case.

	V_T	$m=1$	$m=2$	$m=3$	drift inflation
BTC	1.42	1.20	1.14	1.13	+0.12
ETH	1.37	1.24	1.26	1.29	+0.08
BNB	1.43	1.35	1.36	1.30	+0.13
SOL	1.31	1.15	1.13	1.10	+0.03

drift, and db3 a quadratic one [42, 56]. On synthetic series with ν known by construction all four estimators agree to within 0.01, and an injected ramp inflates the block estimator while leaving the three wavelet estimators fixed — so the comparison has the sensitivity the test requires.

On the data they do not agree, and they disagree the same way in every asset (Table II): the block estimator is the highest of the four throughout, by 0.13 to 0.29. Injecting each asset's own measured mean drift onto a clean synthetic series of $\nu = 1.38$ inflates the block estimate by +0.03 to +0.13 and moves the Haar estimate by less than 0.01, and removing that drift from the data by hand brings the block estimate down from 1.42 to 1.09 in BTC. The gap is the drift.

What this leaves standing. One result survives and two stronger readings do not. The robust result is $\nu > 1$ in all four assets by all four estimators, ranging over 1.10–1.43. Superdiffusive growth of the counting variance is a statement about memory alone — it follows from a correlation decaying more slowly than τ^{-1} — and it is robust to the confounder by construction, since removing the mean cannot manufacture it. What does not survive is a universal two-regime closure: BNB falls below its asset-specific consistency band, and a fit without amplitudes cannot establish quantitative agreement in the other assets.

The second result that does not survive is the drift of ν_{loc} as an established result. It is exhibited only by the estimator we have shown to be susceptible, and the drift-blind estimators are underpowered exactly where it is claimed: the standard error on their local slope grows from ± 0.05 at $T = 8$ to ± 0.22 at $T = 256$, against a claimed excursion of 0.28. They therefore neither reproduce the crossover nor exclude it. We accordingly state Eq. (20) as a prediction that the data are *consistent with but do not discriminate* at this sample size, and we withdraw the inference from it against a single-component bath: a constant ν_{loc} combined with a drifting mean would produce what we observe. Discriminating would need either a longer record or an asset with a stationary mean flow, and we regard it as the sharpest open test the

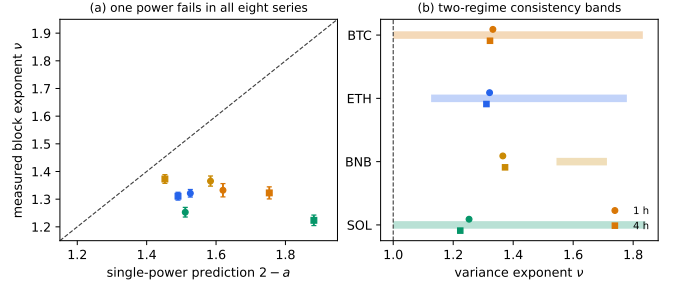


FIG. 4. Scaling reassessment. (a) The broad-window single-power prediction misses all eight series. Circles and squares denote hourly and four-hourly data. (b) Asset-specific bands implied by the two fitted decay regimes; BNB lies outside. The bands omit amplitudes and crossover uncertainty and are therefore consistency checks rather than quantitative predictions.

framework offers.

VII. A MINIMAL TRANSPORT DEVICE

The field theory above says nothing about the counting sector beyond second order, by construction. To probe it we turn to an explicit open-system device whose transport can be computed without approximation, and ask how much of the measured phenomenology it can carry. The device is a single level between two leads with a bias applied as a quench, solved with nonequilibrium Green functions [57]; the counting statistics follow the standard Levitov–Lesovik construction [51, 58]. Computations use an independent solver [59].

A. Setup and numerical controls

We take $\varepsilon_0 = 0$, temperature 0.05, bias $V = 0.3$, and a structured hybridisation combining a wide Lorentzian ($\Gamma = 0.5$, width 4.0) with a narrow slow band ($\Gamma = 0.2$, width 0.04), on a grid of 2561 points with $dt = 0.125$ over $t \in [-120, 200]$ relative to the quench. Three controls are worth quoting because they bound the numerics. The relaxation rate recovered from the unbiased equilibrium run agrees with the analytic total width to 0.33% (-0.9967 against 1.0). Unitarity of the scattering description holds to 8×10^{-16} . The zero-frequency noise and the second counting cumulant, computed by independent routes, agree to 1.3×10^{-9} . Discretisation is the binding constraint: the wide-band product must satisfy $W dt \lesssim 0.5$ or the trapezoidal rule inflates the damping measurably.

B. What the device reproduces

The measurement [12] found a split that is unusual enough to be a sharp target: after a shock the *mean* flow relaxes with an age-dependent profile, while the *correlation* does not age at all. The device reproduces this split exactly, and for a transparent reason.

The mean relaxes in two regimes when — and only when — the bath is structured. With the two-band bath the current after the quench decays with log-slopes of -1.14 ± 0.20 (fast window) and -2.14 ± 0.09 (slow window), an Omori-like exponent of -1.94 ± 0.05 over the intermediate range [Fig. 5(a,b)]. With a single wide Lorentzian the same fits give -3.43 ± 0.12 and $+0.009 \pm 0.001$: one regime, then nothing. The slow band is what buys the second regime.

Meanwhile the correlation does not age. Measuring the correlation time at eight waiting times from $t_w = 10$ to $t_w = 180$, we obtain $\tau_c = 3.375$ at every single one, in all bath configurations, constant to the grid resolution. The reason is structural rather than numerical: with a static Hamiltonian the retarded Green function is rigidly time-translation-invariant after the quench, so the coherence carries no clock even while the occupation is still relaxing.

This is a satisfying result, and we want to state precisely what it does and does not establish. It shows that a bath with memory suffices to produce an ageing mean alongside a non-ageing correlation — no kinetics of model parameters, no genuine ageing of the kernel, is required. It does not show that the market’s particular relaxation shape is reproduced; see Sec. VII D.

C. Where the device fails, and why that is informative

The counting sector is a different story. Here the device is a calibrated benchmark, not a universal exclusion theorem for the market.

Across all bath configurations the device gives $\mathcal{F} = 0.332\text{--}0.341$ under the conventional mesoscopic ratio $\kappa_2/|\kappa_1|$, and a counting variance whose local log-slope runs from 0.94 to 0.99. These are the expected Pauli-antibunched and diffusive values for this finite-memory, single-channel reference.

The market normalisation $\text{Var}(Q_T)/(\bar{q}^2 \langle N_T \rangle)$ uses heterogeneous classical transfers and is not numerically interchangeable with $\kappa_2/|\kappa_1|$. We therefore do not use the difference in absolute levels as an exclusion. The controlled comparison is structural: the calibrated device is diffusive, whereas every drift-resistant market estimator gives $\nu = 1.10\text{--}1.36$ and the observed weekly variance exceeds stratified memory-destroying nulls under both raw and normalised flow definitions [Fig. 5(c)]. The model thus supplies an explicit short-memory benchmark; the empirical attribution to memory comes from the null and from Eq. (17), not from importing a fermionic Fano bound into a classical marked process.

This distinction also localises what the Gaussian device omits. Broad marks can raise the absolute market Fano level without any temporal dependence. The remaining observed/null excess and superdiffusive exponent require persistent organisation of those marks and signs, naturally supplied by interactions between order flow and available liquidity and consistent with the microscopic order-splitting account [5–7].

D. A design bound on transient memory

Attempting to push the device’s slow relaxation toward the measured Omori exponent uncovered a constraint that we think has independent interest.

Sweeping a single slow mode, the transient amplitude obeys a clean power law, $A \propto \gamma^{1.003} W^{0.050}$ with an rms residual of 0.061 in the log fit: linear in the coupling and essentially independent of the mode width. Building an ensemble of slow widths designed to produce a power-law tail with exponent 0.55, the achieved relaxation log-slope is -2.49 ± 0.10 , far steeper than the design target and far steeper than the measured ≈ -0.6 .

The natural suspicion is thermal smearing, so we repeated the ensemble run at a tenfold lower temperature (0.005 instead of 0.05, reducing πT from 0.157 to 0.016). The slope became -2.54 ± 0.02 : unchanged, if anything marginally steeper. *The bound is not thermal.* The consistent interpretation is dephasing by the bias window itself, $V/2 = 0.15$, which was held fixed across both runs and which sets a floor on how slowly a biased channel can relax regardless of how the bath is engineered. The testable consequence is that reaching slower transients requires $V \lesssim 2W_{\min}$, i.e. biases an order of magnitude smaller than those used here.

VIII. INTEGRATED DISCUSSION

The measurements and the model form one argument, but not one undifferentiated claim. The robust empirical statements are superdiffusive counting variance and weekly amplification over stratified temporal nulls under both raw and normalised flow. The sign-magnitude decomposition attributes the larger share to sign persistence in all four primary and all four post-hoc assets. The absolute Fano level is reported only with its declared proxy normalisation, the predeclared all-assets gate is explicitly failed, and calendar sensitivity and cross-asset dependence limit universality claims. The response and second-order counting closure are consistency checks: once the correlation kernel is fixed, they carry no independent second-order evidence.

The exactly solved conductor supplies a calibrated benchmark. It can produce a relaxing mean without ageing its correlation, but its counting statistics remain sub-Poissonian and diffusive under the device’s conventional normalisation. Thus the comparison neither iden-

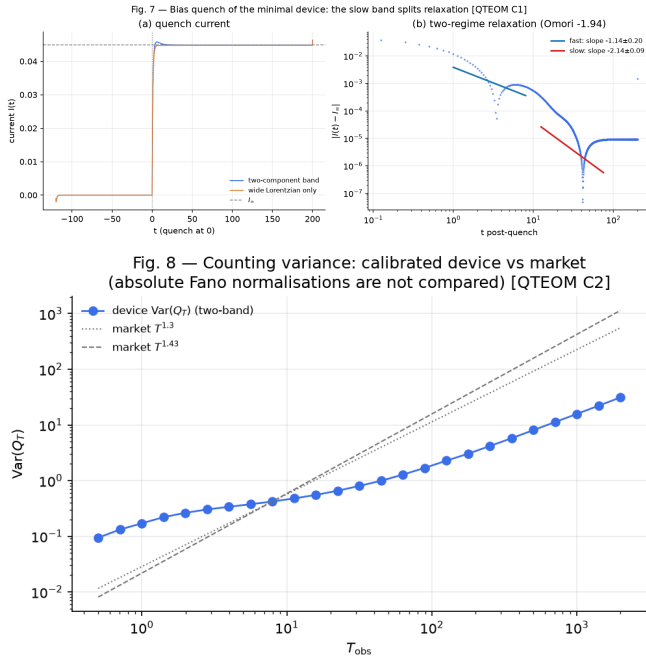


FIG. 5. The minimal device. (a) Current through the bias quench. (b) Relaxation of $|I - I_\infty|$, showing the two regimes that the structured bath produces; a single wide band gives only one. (c) Counting variance of the device against the growth band measured in the market. The device is diffusive; the market is not.

tifies a unique microscopic market model nor transfers a fermionic bound to heterogeneous classical trades. It shows concretely what a finite-memory reference does; the market's departure is established by its stratified temporal nulls. A marked flow process with long memory is the minimal direction left open by the data.

The local drift of the counting exponent is deliberately not promoted to a discovery: the block estimator is sensitive to the measured drift of the mean, while drift-blind estimators are not decisive at the longest windows. Longer records or assets with a stationary mean flow are needed to separate the two mechanisms. This separation of established results, consistency checks, and unresolved tests is the intended scope of the article.

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- [1] M. Ramezani, F. Rahnama Roodposhti, G. Abaspour Esfeden, and M. Ramezani, [Novel market temperature definition through fluctuation theorem: A statistical physics framework for financial crisis prediction](#) (2025), [arXiv:2509.23692 \[cond-mat.stat-mech\]](#).
 - [2] H. Li, Y. Xiao, M. Polukarov, and C. Ventre, *Entropy* **26**, 24 (2024).
 - [3] J. Gao, L. Zhang, P. Fang, P. Ke, J. Wu, Y. Liu, and H. Zhou, [arXiv preprint 2604.14962](#) (2026).
 - [4] Y. Yura, H. Takayasu, D. Sornette, and M. Takayasu, *Phys. Rev. Lett.* **112**, 098703 (2014).
 - [5] F. Lillo, S. Mike, and J. D. Farmer, *Phys. Rev. E* **71**, 066122 (2005).
 - [6] Y. Sato and K. Kanazawa, *Phys. Rev. Lett.* **131**, 197401 (2023).
 - [7] Y. Sato and K. Kanazawa, *Phys. Rev. Research* **5**, 043131 (2023).
 - [8] D. E. Taranto, G. Bormetti, J.-P. Bouchaud, F. Lillo, and B. Tóth, *Quantitative Finance* **18**, 903 (2018), [arXiv:1602.02735](#).
 - [9] G. Maitrier and J.-P. Bouchaud, *Quant. Finance* **10.1080/14697688.2026.2641766** (2026), [arXiv:2506.07711](#).
 - [10] F. Lillo and J. D. Farmer, *Stud. Nonlinear Dyn. Econom.* **8**, 10.2202/1558-3708.1226 (2004).
 - [11] F. Shi, N. Aden, S. Huang, N. Johnson, X. Sun, J. Gao, L. Xu, H. Shen, X. Cheng, and C. Song, [arXiv preprint](#) (2021), [arXiv:2101.06236 \[q-fin.TR\]](#).
 - [12] N. Bolívar, G. Abellán, and D. Verrilli, [keldysh-finance: transport analysis of signed order flow](#) (2026), analysis code, data provenance, and deposited numerical outputs.
 - [13] M. Naviglio, G. Bormetti, F. Campigli, G. Rodikov, and F. Lillo, [Why is the estimation of metaorder impact with public market data so challenging?](#) (2025), [arXiv:2501.17096 \[q-fin.TR\]](#).
 - [14] A. Gerig, [A theory for market impact: How order flow affects stock price](#) (2008), [arXiv:0804.3818 \[q-fin.ST\]](#).
 - [15] T. Jaisson, [Market impact as anticipation of the order flow imbalance](#) (2014), preprint; no journal version located, [arXiv:1402.1288 \[q-fin.TR\]](#).
 - [16] J. Muhle-Karbe, Y. Ouazzani Chahdi, M. Rosenbaum, and G. Szymanski, [A unified theory of order flow, market impact, and volatility](#) (2026), [arXiv:2601.23172 \[q-fin.TR\]](#).
 - [17] D. Andrieux and P. Gaspard, *Journal of Statistical Physics* **127**, 107 (2007).
 - [18] A. C. Barato and R. Chetrite, *Journal of Physics A: Mathematical and Theoretical* **45**, 485002 (2012).
 - [19] D. Andrieux and P. Gaspard, *Journal of Statistical Mechanics: Theory and Experiment* **2007**, P02006 (2007).
 - [20] Supplemental material for “sign memory amplifies counting noise in market order flow” (2026), higher cumulants, sign-magnitude nulls, bootstrap uncertainty, stratification sensitivity, and scaling controls.
 - [21] F. Lillo and R. N. Mantegna, *Physical Review E* **68**, 016119 (2003).

- [22] A. G. Zawadowski, G. Andor, and J. Kertész, *Quantitative Finance* **6**, 283 (2006).
- [23] F. Selçuk, *Physica A* **333**, 306 (2004).
- [24] K. Pearson, *Proceedings of the Royal Society of London* **60**, 489 (1897).
- [25] R. A. Kronmal, *Journal of the Royal Statistical Society A* **156**, 379 (1993).
- [26] A. Ciuparu and R. C. Mureşan, *European Journal of Neuroscience* **43**, 861 (2016).
- [27] K. J. Forbes and R. Rigobon, *The Journal of Finance* **57**, 2223 (2002).
- [28] B. H. Boyer, M. S. Gibson, and M. Loretan, *Pitfalls in tests for changes in correlations*, International Finance Discussion Paper 597 (Board of Governors of the Federal Reserve System, 1999).
- [29] J. J. Brey, A. Prados, M. I. García de Soria, and P. Maynar, *Journal of Physics A: Mathematical and Theoretical* **40**, 14331 (2007).
- [30] M. G. Kendall, *Biometrika* **41**, 403 (1954).
- [31] R. Zeraati, T. A. Engel, and A. Levina, *Nature Computational Science* **2**, 193 (2022).
- [32] T. Akimoto, E. Yamamoto, and T. Uneyama, *Journal of the Physical Society of Japan* **92**, 074005 (2023).
- [33] S. Abe and N. Suzuki, *Physica A* **332**, 533 (2004).
- [34] V. Usmanova, Y. V. Lysogorskiy, and S. Abe, *Europhysics Letters* **121**, 48001 (2018).
- [35] F. X. Diebold and R. S. Mariano, *J. Bus. Econ. Stat.* **13**, 253 (1995).
- [36] W. K. Newey and K. D. West, *Econometrica* **55**, 703 (1987).
- [37] P. H. Westfall and S. S. Young, *Resampling-Based Multiple Testing: Examples and Methods for p-Value Adjustment* (Wiley, 1993).
- [38] International Council for Harmonisation, *ICH Harmonised Tripartite Guideline E10: Choice of Control Group and Related Issues in Clinical Trials*, Tech. Rep. (ICH, 2000) step 4; CPMP/ICH/364/96.
- [39] J. M. Hoenig and D. M. Heisey, *The American Statistician* **55**, 19 (2001).
- [40] P. Abry and D. Veitch, *IEEE Transactions on Information Theory* **44**, 2 (1998).
- [41] D. Veitch and P. Abry, *IEEE Transactions on Information Theory* **45**, 878 (1999).
- [42] I. Daubechies, *Ten Lectures on Wavelets* (SIAM, Philadelphia, 1992).
- [43] P. Flandrin, *IEEE Transactions on Information Theory* **38**, 910 (1992).
- [44] R. B. Davies and D. S. Harte, *Biometrika* **74**, 95 (1987).
- [45] H. Wold, *A Study in the Analysis of Stationary Time Series* (Almqvist and Wiksell, 1938).
- [46] J. D. Hamilton, *Time Series Analysis* (Princeton University Press, 1994).
- [47] J.-P. Bouchaud, Y. Gefen, M. Potters, and M. Wyart, *Quant. Finance* **4**, 176 (2004).
- [48] J.-P. Bouchaud, J. Bonart, J. Donier, and M. Gould, *Trades, Quotes and Prices: Financial Markets Under the Microscope* (Cambridge University Press, 2018).
- [49] L. F. Cugliandolo, J. Kurchan, and L. Peliti, *Phys. Rev. E* **55**, 3898 (1997).
- [50] L. S. Levitov and G. B. Lesovik, *JETP Lett.* **58**, 230 (1993).
- [51] L. S. Levitov, H. Lee, and G. B. Lesovik, *J. Math. Phys.* **37**, 4845 (1996).
- [52] K. Schönhammer, *Phys. Rev. B* **75**, 205329 (2007).
- [53] M. Esposito, U. Harbola, and S. Mukamel, *Rev. Mod. Phys.* **81**, 1665 (2009).
- [54] G. Gallavotti and E. G. D. Cohen, *Phys. Rev. Lett.* **74**, 2694 (1995).
- [55] J. L. Lebowitz and H. Spohn, *J. Stat. Phys.* **95**, 333 (1999).
- [56] D. B. Percival and A. T. Walden, *Wavelet Methods for Time Series Analysis* (Cambridge University Press, Cambridge, 2000).
- [57] A.-P. Jauho, N. S. Wingreen, and Y. Meir, *Phys. Rev. B* **50**, 5528 (1994).
- [58] Y. M. Blanter and M. Büttiker, *Phys. Rep.* **336**, 1 (2000).
- [59] Quantumtransporteom: equation-of-motion solver for nonequilibrium quantum transport (2026), version 0.3.0.